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Bearing fault diagnosis and prognosis using data fusion based feature extraction and feature selection

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ABSTRACT

The extraction of significant features is essential for efficient fault diagnosis and prognosis of rolling element bearing. Data fusion is the predominant technology for extracting significant features by fusing several original features. In this paper, seventy-two original features are extracted from bearing vibration data using various signal processing techniques. The relevant features subset is selected from the extracted features using the Random Forest method. The selected features are fused by fourteen dimensionality reduction techniques to extract 2D fault features and health indicators, and a comparison is made between the techniques to identify the most efficient technique. The Bhattacharyya distance and Support vector machine are used to verify fault diagnosis accuracy. A new index is computed for selecting the suitable prognosis health indicator, and the Long short-term memory technique is used to predict the remaining useful life of bearing. Two real-world bearing datasets are utilized to validate the proposed methodology.

1. Introduction

Rolling element bearing is a critical component in rotating machinery carrying dynamic loads. More than 90 percent of rotating machines use bearings to support the machine structure and allow rotational motion [1]. The sudden bearing breakdown may cause catastrophic failure of machines resulting in loss of production, increased downtime, and higher risk for human safety. More than 40 percent of industrial motors fail due to bearing failure. Therefore, fault diagnosis and prognosis play a significant role in identifying incipient faults and forecasting the upcoming faults in the bearing to avoid catastrophic failure reducing productivity losses and downtime while increasing human safety [2,3]. To expand reliability, maintainability, and safety requirements, constructing efficient fault diagnosis and prognosis models are essential.

Fault diagnosis identifies the early fault occurrence, location of the fault, and severity of the fault by which malfunctioning of machines may be prevented [4]. The fault diagnosis models are primarily dependent on the fault features extracted from the condition monitoring data. Prognosis is the process of forecasting the degradation trend and predicting the Remaining Useful Life (RUL) of the machines [5]. There are various physical-based, data-driven, and hybrid prognosis models available in the literature. The physical-based models formulate a mathematical equation based on the physical degradation process to forecast the RUL of the machines. These models are defect and component-specific. The conceptual understanding of the degradation process is essential for model building. Therefore, it is challenging to construct mathematical equations for complex systems and processes.

Data-driven models require condition monitoring data to build the prognosis model for characterizing defect progression and RUL prediction [6]. These models work independently of the physical degradation process and are better in precision and complexity. Therefore, extracting the maximum available information as fault features from condition monitoring data and constructing efficient health indicators are essential for constructing the fault diagnosis and data-driven prognosis models. With the recent increase in computing efficiency and sample data storage, machine learning techniques are being applied in many fault diagnosis and prognosis applications.

Zhao and Jia [7] proposed K-Nearest Neighbors (KNN) based fault diagnosis framework for bearing fault diagnosis. The authors extracted significant lower-dimensional features using global–local marginal Fisher analysis, and these features are given as input to the Euclidean weighted KNN method to diagnose bearing faults. Choudhary et al. [8] used the Convolutional Neural Network (CNN) method for bearing fault diagnosis. In this work, the authors utilized thermal images as input to the CNN algorithm for diagnosing the faults. The results were compared with the Artificial neural network and showed their superiority. The Support Vector Machine (SVM) is the most successfully used classification technique in literature due to its generalization capability. Wang et al. [9] developed Optimized-SVM for bearing fault diagnosis.

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List of Abbreviations

BD	Bhattacharyya Distance
BPFI	Ball Pass Frequency Inner Race
BPFO	Ball Pass Frequency Outer Race
BSF	Ball Spin Frequency
CCA	Canonical Correlation Analysis
CNN	Convolutional Neural Network
CWT	Continuous Wavelet Transform
DM	Diffusion Maps
DWT	Discrete Wavelet Transform
EMD	Empirical Mode Decomposition
FA	Factor Analysis
FFT	Fast Fourier Transform
FTF	Fundamental Train Frequency
HFRT	High-Frequency Resonance Technique
ICA	Independent Component Analysis
ISOMAP	Isometric Mapping
KNN	K-Nearest Neighbors
LDA	Linear Discriminant Analysis
LE	Laplacian Eigenmaps
LLE	Locally Linear Embedding
LPP	Locality Preserving Projections
LSTM	Long Short-Term Memory
LTSA	Local Tangent Space Alignment
MAE	Mean Absolute Error
MD	Mahalanobis Distance
PCA	Principal Component Analysis
RBF	Radial Basis Function
RMS	Root Mean Square
RMSE	Root Mean Square Error
SE	Shannon Entropy
SNR	Signal-to-Noise Ratio
STD	Standard Deviation
SV	Singular Values
SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighborhood Embedding

The supervised Isometric Mapping (ISOMAP) was used to extract the lower-dimensional data from original data, and these features were fed in the Grasshopper Optimization-SVM method to classify various bearing faults. Mahata et al. [10] designed a fault diagnosis framework for grinding wheel wear identification using Hilbert Huang Transform and the SVM. The Long Short-Term Memory (LSTM) is a predominant technique to analyze time-series data due to its handling capacity of long-term dependencies in time series data. Ding et al. [11] applied an LSTM-based prognosis method for journal bearing. The authors used fuzzy c-means techniques to find various degradation stages, and Particle swarm optimization was used to tune the hyper-parameters of the LSTM to predict the bearing RUL.

The real-world industrial machines generate non-linear and non-stationary signals. The vibration signals acquired from rotating machines are widely used for condition monitoring due to their high sensitivity to defects [12,13]. The features extracted from the bearing vibration signal may be used for fault diagnosis and prognosis. However, the traditional features may not be sensitive to early defects; and, in the later stage, may have considerable fluctuations in the degradation trend [14]. These features may not contain complete information regarding the health state of bearings. Therefore, applying these features directly in fault diagnosis and prognosis may produce less accurate results. Merging the multiple traditional features extracted from the raw vibration signal to a single feature by data fusion may increase the sensible content of the health state information resulting in high accuracy of the bearing fault diagnosis and prognosis. Data fusion is the process of integrating several original features into more significant meaningful features. A large number of features extracted from vibration signals may cover the vast information. However, redundant information may also be submerged in it, reducing the model accuracy; increasing complexity and computational time. Therefore, extracting useful information from the high-dimensional feature set by eliminating redundant information is crucial. The dimensionality reduction techniques have applications in machinery fault diagnosis and prognosis to fuse the high dimensional feature set to significant lower-dimensional features by eliminating redundant information and noise. Fusing the higher-dimensional feature set directly by data fusion techniques requires more time and leads to overfitting and the curse of dimensionality problems. The selection of relevant features subset before fusion is essential to reduce the complexity and overfitting. The literature has several feature selection methods to select the essential features. Traditionally, filter-based [15], wrapper-based [16], and embedded methods [17] are used for feature selection.

Researchers have used different linear and non-linear dimensionality reduction techniques as data fusion techniques to extract significant features by fusing a high dimensional feature set. Data fusion is advantageous in initial faults detection, classification of different faults types, and constructing a health indicator for RUL prediction. The Principal Component Analysis (PCA) is a popularly used traditional linear dimensionality reduction technique representing the maximum variance in the data. Zvokelj et al. [18] developed a fault diagnosis framework by integrating Ensemble empirical mode decomposition and the PCA for bearing fault diagnosis. The bearing signal was decomposed into different time scales using Ensemble empirical mode decomposition. These decomposed data were given input to multiscale PCA for determining incipient faults in the bearing. Wang et al. [19] computed a health indicator using the PCA method by fusing various statistical features. A novel life prediction model was developed using the PCA-based health indicator to find the bearing RUL. Independent Component Analysis (ICA) may be used for effective fault diagnosis and noise filtering in the data. Li et al. [20] combined compression sensing and the ICA to extract fault features from multi-source mixed data for bearing fault diagnosis.

Shakya et al. [21] established a health indicator using Mahalanobis Distance (MD) for bearing health monitoring and used it for online detection of various damage stages in naturally progressing bearing defects [22]. Soylemezoglu et al. [23] followed the MD-based methodology for fault clustering and RUL prediction of the bearing. The MD-based threshold was formulated for fault detection and isolation. After detecting the fault, a linear approximation method was used to find the bearing RUL. Buchaiah and Shakya [24] used the PCA, the ICA, and the MD for bearing fault detection and health monitoring, and a comparison was made between them to find their effectiveness in early fault detection and fault states representation. Dong et al. [25] introduced a methodology based on Factor Analysis (FA) and gene expression programming for transformer fault classification. A lower dimensional data extracted from the FA was used as input to the gene expression programming classifier. Li et al. [26] employed Canonical Correlation Analysis (CCA) (also called Canonical variates analysis) method for fault detection and identification in a centrifugal compressor. Later, the CCA is combined with the LSTM to predict the system condition after fault occurrence.

Linear Discriminant Analysis (LDA) determines the projection direction maximizing the distance between the data clusters commonly used for fault discrimination. Zhou et al. [27] extracted various statistical features from Transient-extracting transform applied to the bearing signal. These features were fused by the LDA algorithm and extracted

3D features for bearing fault classification. Locality Preserving Projections (LPP) is a linear projection map that effectively preserves the local intrinsic geometry of the data in lower-dimensional feature space. Zheng et al. [28] formulated a methodology by integrating transfer learning and LPP for intelligent fault detection of rotating machinery. The transfer LPP method was trained with source data obtained from the variable operational conditions, and this trained knowledge was used to classify the faults in the new target data.

The linear dimensionality reduction techniques may analyze linear data sets effectively. However, they may not extract non-linear information from the non-linear data. Therefore, the non-linear dimensionality reduction techniques have been exploited for analyzing real-world industry machine health states. Locally Linear Embedding (LLE) is a classical non-linear dimensionality reduction technique that uses linear transformations to extract non-linear information. Shao et al. [29] derived a feature index using the LLE method to quantify the bearing degradation trend. An optimized deep belief network-based methodology was constructed using the formulated index as input to detect the bearing faults and predict RUL. Laplacian Eigenmaps (LE) uses a spectral technique to map the high-dimensional data into lower-dimensional feature space. Wang et al. [30] exploited the LE technique to compute an early warning and good monotonic health indicator by fusing various original features extracted from the bearing vibration signal using the graph spectrum reconstruction method. Diffusion Maps (DM) retains the global structure of the data in lower-dimensional feature space by preserving as much of the pairwise diffusion distance in the data. Sipola et al. [31] constructed a DM-based framework to extract lower-dimensional features used for gear classification. New data were classified as working or failure gears in the lower-dimensional space, and warning levels were also identified.

ISOMAP extracts significant feature space from observed data by retaining local neighborhood information. Su et al. [32] computed 2D fault features using the ISOMAP by compressing the high-dimensional features, and these fault features were given as input to kernel-based extreme learning machine to detect the bearing faults automatically. Local Tangent Space Alignment (LTSA) uses locally linear transformations to map observed data into lower-dimensional space. Chen et al. [33] applied the LTSA to extract high sensitive 3D fault features by fusing various traditional features, and these 3D features were input to the optimized SVM for classifying bearing faults. The t-Distributed Stochastic Neighborhood Embedding (t-SNE) is an unsupervised dimensionality reduction technique, which uses conditional probability-based similarity measurement between the data points. Zheng et al. [34] fused various features using the t-SNE to extract significant 2D and 3D features. Then, these features were used as input to the variable predictive models-based class discrimination method for classifying various types and severity of bearing faults.

Selecting a suitable dimensionality reduction technique for fault classification and prognosis of different machinery components is a challenging task. As per the best of the authors' knowledge, no work has been reported in the literature on comparing dimensionality reduction techniques for bearing fault diagnosis and prognosis. In this work, fourteen dimensionality reduction techniques are performed to extract 2D fault features used for fault classification and computed health indicators for RUL prediction of the bearing. The best techniques for fault classification and RUL prediction have been identified by comparing these fourteen techniques. First, a subset of features is selected from the original feature set using the Random Forest method. The authors have demonstrated that the choice of discriminating features reduces the processing time and overfitting. The selected features subset is fused by the proposed dimensionality reduction techniques and extracted 2D fault features to classify single faults and combined faults. A multi-class SVM method is used to classify the various bearing faults, and classification accuracy is computed. A Bhattacharyya distance-based index is calculated to determine the capability of separating faults. The SVM-based classification accuracy and the Bhattacharyya distance are used to identify the best dimensionality techniques for bearing fault classification. A fitness value is computed by the weighted sum of four attributes (monotonicity, prognosability, trendability, and signal-to-noise ratio) of fourteen health indicators used to select the best health indicator. Finally, the health indicators chosen are utilized to construct the LSTM-based prognosis model for RUL prediction of the bearing. The block diagram of the proposed methodology is presented in Fig. 1.

The major contributions of the proposed work are as follows:

1. Seventy-two original features are derived from bearing vibration signals using multiple signal processing techniques. The best subset of features is selected by the Random Forest method to minimize time consumption and eliminate overfitting.
2. The selected subset of features are fused by fourteen data fusion techniques, and the extracted 2D fault features are used to classify single faults and combined faults and constructed a health indicator for bearing RUL prediction.
3. The obtained 2D fault features are used to calculate the Bhattacharyya distance between fault clusters to measure the capability of separating faults for the various techniques. The 2D fault features are given as input to the SVM to calculate classification accuracy. The separation capability index and classification accuracy are used to compare the fourteen techniques, and the best technique for fault classification is identified.
4. The best health indicator is selected by computing the fitness value. A selected health indicator is used to build the LSTM-based prognosis model for predicting bearing RUL.

The following section provides a brief introduction to feature extraction and selection methods. Section 3 presents a brief introduction to data fusion techniques and their advantages and disadvantages. Section 4 reports the fault diagnosis and prognosis methods. Section 5 provides the details of experimental rigs. Section 6 reports the results and discussion. The major conclusions of the work are discussed in Section 7.

2. Brief introduction on feature extraction and feature selection methods

2.1. Signal processing

Vibration-based condition monitoring methods are primarily used in fault diagnosis and prognosis due to the high sensitivity of the accelerometer. The features extracted from the vibration signals contain health state information of machine components and may play a vital role in fault diagnosis and prognosis. The signal processing techniques in various domains (i.e., time, frequency, and time–frequency domain) have been applied to the acquired vibration data to extract various original features. The time-domain analysis is the simplest technique used in the early stage of mechanical fault diagnosis, which provides the amplitude information of the signal concerning time. However, it misses the frequency-related information in the signal. In the frequency domain analysis, frequency-related information can be extracted, providing information on defect frequencies of the components. The frequency-domain analysis is advantageous in defect type identification in bearings. However, it misses the time information. Therefore, the time–frequency domain analysis is utilized to extract features where both time and frequency information may be obtained, more effective in the mechanical faults diagnosis and prognosis.

In the present work, vibration data acquired from the rolling element bearing is used for condition monitoring. First, the vibration data is filtered with an Empirical Mode Decomposition (EMD) based filter [35], and noise is eliminated. Then, seventy-two features are extracted from bearing vibration data. Among these ten are statistical features (clearance factor, crest factor, impulse factor, peak factor, kurtosis, Root Mean Square (RMS), RMS*kurtosis, shape factor, skewness,

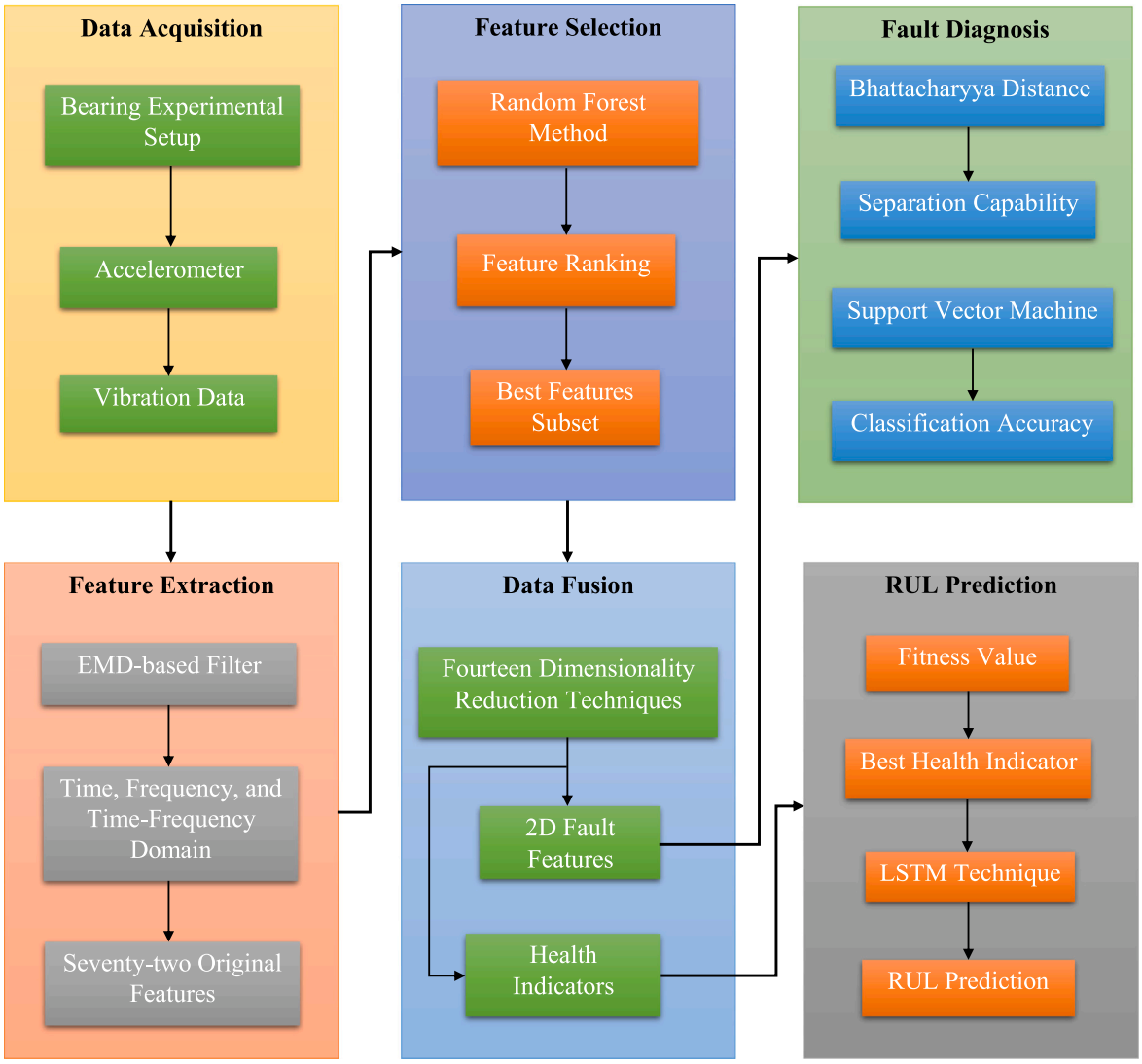


Fig. 1. The block diagram of the proposed methodology.

and Standard Deviation (STD)) extracted in the time-domain. In the frequency domain, eight features are extracted from Fast Fourier Transform (FFT) and High-Frequency Resonance Technique (HFRT) (amplitude at defect frequencies: Ball Pass Frequency Inner Race (BPFI), Ball Pass Frequency Outer Race (BPFO), Ball Spin Frequency (BSF), and Fundamental Train Frequency (FTF)). In the time–frequency domain, twenty-four features are extracted from each Continuous Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT) (Nine statistical features, four features of amplitudes at bearing defect frequencies in each FFT and HFRT, five quantiles, Acosh, and Asinh), and six feature from the EMD (Singular Values (SV) and Shannon Entropy (SE) of the first three intrinsic mode functions). The MATLAB®is used for performing signal processing techniques. The seventy-two original features extracted from the bearing vibration signal in all three domains are given in Table 1.

2.2. Random forest

The high dimensional feature set extracted by the aforementioned techniques may contain redundant features resulting in overfitting, complexity, and high computational time. The best subset of lower-dimensional feature selection from the initial feature set is essential to eliminate these problems. Several feature selection methods (filter-based methods, wrapper-based methods, and embedded methods) are available in the literature to select relevant features. In this case, the Random Forest method (an embedded feature selection method) is used to determine the best subset of lower-dimensional features. The Random Forest combines multiple decision trees, each working as a classifier, and each tree is sampled from the input feature set to build a feature subset [36]. Each tree consists of three nodes: root node, leaf node, and decision node. In the decision tree, root nodes are split into decision nodes, and decisions nodes are branched as leaf nodes. The final output of each tree is obtained at a leaf node. The majority voting process selects the final features subset. The Random Forest network diagram is represented in Fig. 2.

The Random Forest method ranks the features based on the Gini index or information gain for classification and the variance reduction for regression problems. The Gini index is calculated by the following formula.

$$Gini\ index = \sum_{i=1}^{C} P_i(1 - P_i) \tag{1}$$

where P_i is the proportion of data points belonging to class i for a particular node, and C is the total number of classes.

First, the Random Forest is processed to rank the features, and the top-ranked features are selected as a subset. Then, the Random Forest model accuracy of the subset is determined and compared with the model accuracy with total features. The procedure is repeated to select

Table 1
Original features extracted in various domains.

Domain	Features					
Time	RMS	Crest factor	Peak factor	Kurtosis	Clearance factor	RMS*Kurtosis
	Skewness	Impulse factor	Shape factor	STD		
Frequency	FFT_{BPFI}	FFT_{BPFO}	FFT_{BSF}	FFT_{FTF}	$HFRT_{BPFI}$	$HFRT_{BPFO}$
	$HFRT_{BSF}$	$HFRT_{FTF}$				
Time-frequency	CWT	Clearance factor	Crest factor	Peak factor	Kurtosis	RMS
		RMS*Kurtosis	Shape factor	Skewness	Impulse factor	FFT_{BPFI}
		FFT_{BPFO}	FFT_{BSF}	FFT_{FTF}	$HFRT_{BPFI}$	$HFRT_{BPFO}$
		$HFRT_{BSF}$	$HFRT_{FTF}$	Asinh	Acosh	Quantile(0)
		Quantile(0.25)	Quantile(0.5)	Quantile(0.75)	Quantile(1)	
	DWT	Clearance factor	Crest factor	Peak factor	Kurtosis	RMS
		RMS*Kurtosis	Shape factor	Skewness	Impulse factor	FFT_{BPFI}
		FFT_{BPFO}	FFT_{BSF}	FFT_{FTF}	$HFRT_{BPFI}$	$HFRT_{BPFO}$
		$HFRT_{BSF}$	$HFRT_{FTF}$	Asinh	Acosh	Quantile(0)
		Quantile(0.25)	Quantile(0.5)	Quantile(0.75)	Quantile(1)	
	EMD	SV1	SV2	SV3	SE1	SE2
		SE3				

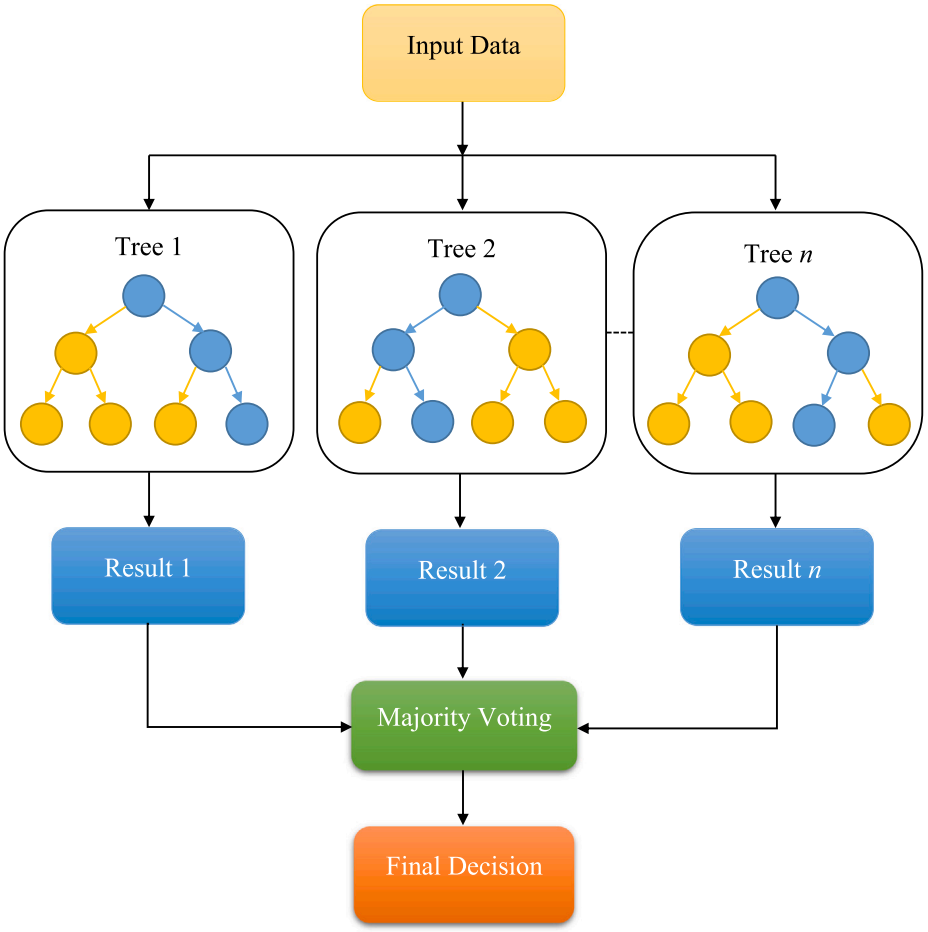


Fig. 2. Random Forest network.

the best subset of features such that it should be the smallest subset with the maximum accuracy. The code to compute the Random Forest algorithm is written in the Python language.

3. Brief introduction on data fusion methods

Dimensionality reduction techniques convert high-dimensional features into lower-dimensional features by linear or non-linear transformations by preserving intrinsic geometry or properties of the original data. The significant advantage of these techniques is to extract useful lower-dimensional features by fusing high-dimensional original features without losing useful information [37]. Fourteen dimensionality reduction techniques are used for fusing the high-dimensional original feature set to significant low-dimensional features (best health indicators and 2D fault features). The conceptual details of the proposed fourteen techniques (1 to 7 are linear and 8 to 14 are non-linear techniques) are presented in Table 2. In this work, the dimensionality reduction techniques are performed using MATLAB®.

Table 2
Conceptual summary of dimensionality reduction techniques.

S. no.	Technique	Process of data transformation	Advantages	Disadvantages	References
1	PCA	Eigenvalue decomposition	Removes the correlation between the features. Improves data visualization.	Limited to linear transformations. It fails in cases where mean and covariance are not enough to define datasets.	[18,19]
2	ICA	Non-Gaussian distribution	Filters noise from the signal.	High training time required.	[20]
3	MD	Orthogonal process	Multivariate outliers can be identified.	Due to instability of the covariance matrix, calculations are unstable.	[21,23]
4	FA	Factor rotation	Identifies the groups of inter-related variables. Hidden dimensions can be identified.	Naming of the factors can be difficult.	[25]
5	LDA	Class separability by scatter matrix	Classifies the data into groups. Simple, fast and portable technique	Suffers from class singularity issue	[27]
6	CCA	Linear combinations	Solves problems using closed-form analytical solution.	Limited to linear transformations.	[26]
7	LPP	Transfer matrix	Preserves the local structure of the data. Discovers the nonlinear properties of the data also.	Does not consider label information. Graph construction of LPP is sensitive to noise and outliers.	[28]
8	KPCA	Kernel mapping	Better capability in non-linear projections.	High training time required.	[38]
9	ISOMAP	Neighborhood graph	Preserves "true" relationship between data points.	Suffers from topological instability.	[32]
10	LLE	Reconstruction weights	Preserves local geometry. Less computational time.	Sensitivity to noise. Consumes more memory.	[29]
11	LE	Spectral technique	Provides unique solution.	May produce disconnected neighborhood graph.	[30,39]
12	DM	Markov random walk	Robustness to noise and small perturbations of the data.	Limitation in detecting small anomalies.	[31]
13	LTSA	Tangent space alignment	Preserve local properties.	Ignores the label information.	[33]
14	t-SNE	Conditional probabilities	Efficiently handles the non-linear data.	Output features are limited to 2D or 3D. Computationally Complex.	[34]

4. Fault diagnosis and prognosis methodologies

4.1. Bhattacharyya Distance

Bhattacharyya Distance (BD) is a dimensionless number used to find similarities between two probability distributions and measure the class separability between the clusters in the classification problems [40]. The BD value is high when data clusters are separated well. It may be applied to samples with different standard deviations. The BD value increases when the difference between standard deviations of samples increases. It is also used in feature extraction, feature selection, and classification error calculation.

$$BD = \frac{1}{8}(\mu_2 - \mu_1)^T \left[\frac{C_1 + C_2}{2}\right]^{-1} (\mu_2 - \mu_1) + \frac{1}{2}\ln\frac{|(C_1 + C_2)/2|}{\sqrt{|C_1||C_2|}} \tag{2}$$

where μ_i and C_i are the mean and covariance matrix of ith class samples.

4.2. Support vector machine

The SVM is a supervised learning technique developed by Cores and Vapnik [41], which uses statistical learning theory to analyze the data for classification. It uses structural risk minimization, maximizing the margin between the different clusters. The SVM method is mainly used for fault detection and classification of mechanical components. The SVM has shown many successful results due to its generalization capability. It is developed for binary classification. In this method, the boundary line may be indicated by linear function $w^T x + b$, where x is the feature vector with class labels -1 and 1, w indicates weight vector, and b represents the shift of the separating hyperplane. The main aim of the SVM is to maximize the margin between the classes by separating the hyperline. The optimization problem is defined as:

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \\ \text{subjected to} \quad & y_i(w^T(x_i + b)) \geq 1 - \xi_i \\ & \xi_i \geq 0, \qquad i = 1, 2, \ldots, n \end{aligned} \tag{3}$$

The Lagrangian multiplier technique is used to solve the constrained quadratic problem, and its dual problem is formulated as:

$$\begin{aligned} \max \quad & \sum_{i}^{n}\lambda_i - \frac{1}{2}\sum_{i}^{n}\sum_{j}^{n}\lambda_i\lambda_j y_i y_j K(x_i, x_j) \\ \text{subjected to} \quad & \sum_{i}^{n}\lambda_i y_i = 0, \qquad 0 \leq \lambda_i \leq C, \qquad i = 1, \ldots, n \end{aligned} \tag{4}$$

where λ_i indicates a Lagrangian multiplier and $K(x_i, x_j)$ is a kernel function.

The solution to the SVM for unknown data is given by the equation:

$$f(x) = sign(\sum_{i=1}^{n}\lambda_i y_i K(x_i, x_j) + b), \qquad f(x) \to \{-1, 1\} \tag{5}$$

Many real-world problems exhibit multi-class classification. The multi-class SVM can be formulated by multiple binary SVMs, such as one-to-one and one-to-all approaches. The MATLAB®toolbox is used to perform the SVM in this work.

4.3. Best health indicator selection

Appropriate health indicator selection is essential for building an efficient prognosis model leading to accurate bearing RUL prediction. The significant health indicator should have four qualities: monotonicity, prognosability, trendability, and signal-to-noise ratio. The weighted sum of these four attributes may be used to identify the better prognosis health indicator for RUL prediction. The range of these parameters may vary from zero to one. The value zero represents the low score of a particular parameter, which means the parameter is not suitable for the RUL prediction. The value one indicates the highest score of the parameter, more suitable for the RUL prediction.

4.3.1. Monotonicity

Monotonicity measures the degree of the positive or negative trend of the health indicator. The monotonicity ignores the probability of self-healing of the systems. The importance of monotonicity is its applicability to a specific system. The monotonic trend is valid for an

entire system, even though individual components may undergo self-healing. The monotonicity of the features population is expressed as an average absolute difference of a fraction of positive and negative derivatives for each trend [42].

$$Monotonicity(M) = mean\left(\left|\frac{+ve(\frac{dy}{dx})}{n-1} - \frac{-ve(\frac{dy}{dx})}{n-1}\right|\right) \quad (6)$$

where n indicates the number of observations in a sample.

4.3.2. Prognosability

Prognosability measures the variance in the critical failure value of a population of features. It quantifies the fraction of the variance of the final failure values for each trend and the mean of the trend ranges. It is exponentially weighted to give the desired zero to one scale [42].

$$Prognosability(P) = exp\left(-\frac{std(failure\ values)}{mean(failure\ values - starting\ values)}\right) \quad (7)$$

4.3.3. Trendability

Trendability measures the degree of the population of features of a system with the same underlying shape and expressed by the same functional form. It is computed by comparing the fraction of positive first and second derivatives of each feature in the population. A feature is considered to have good trendability if all features in the population are modeled by the same functional form [42].

$$Trendability(T) = 1 - std\left(\frac{+ve\frac{dy}{dx}}{n-1} + \frac{+ve\frac{d^2y}{dx^2}}{n-2}\right) \quad (8)$$

where n indicates the number of observations in ith feature.

4.3.4. Signal-to-Noise Ratio

Signal-to-Noise Ratio (SNR) is defined as the ratio of the power of the desired signal to the power of the undesired noise [43]. It compares the level of signal to the level of noise. The SNR measures the quality of the signal.

$$SNR = 10log\left(\frac{P_{signal}}{P_{noise}}\right) \quad (9)$$

where P_{signal} is the power of the signal, and P_{noise} is the power of the noise.

The fitness value-based new index (H) is calculated by taking the weighted sum of the above four attributes given by the following equation:

$$H = w_M M + w_P P + w_T T + w_S SNR \quad (10)$$

4.4. Long Short Term Memory

The Long Short Term Memory (LSTM) has been proposed by Hochreiter and Schmidhuber [44] to resolve the vanishing gradient problem in recurrent neural networks, which is unable to handle the long-term dependencies in time series data. The LSTM technique uses long-term dependence (advantageous in time series data analysis) and is an effective tool for time series prediction. The LSTM network mainly contains three gates: forget, input, and output gates. These three gates are connected to decide which information to store or forget. The key part of the LSTM is the cell state (C). The *sigmoid* and *tanh* functions are implemented to select the information. The architecture of the LSTM network is shown in Fig. 3. The LSTM algorithm is computed using MATLAB®in this work.

An input (x_t) and the previous hidden layer output (h_{t-1}) are given to the network at time instant t. The forget gate first decides which information may be discarded or kept, i.e., it filters the important historical information. The output of the forget gate is computed as:

$$f_t = \sigma[U_f h_{t-1} + V_f x_t + b_f] \quad (11)$$

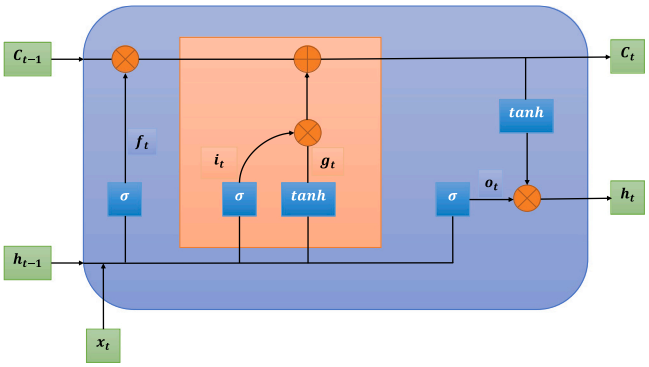


Fig. 3. Architecture of the LSTM technique.

where U_f and V_f are input weights, and b_f is the bias of the forget gate.

Next, inputs x_t and h_{t-1} are passed through the input gate to select the information that will be stored in the cell state. The input gate will decide the updation of states, and the intermediate cell state uses the *tanh* function to create a new vector added to the cell. The input gate and intermediate states cells are given by the following equations:

$$i_t = \sigma[U_i h_{t-1} + V_i x_t + b_i] \quad (12)$$

$$g_t = tanh[U_g h_{t-1} + V_g x_t + b_g] \quad (13)$$

where (U_i, V_i), b_i are input weights, and the bias of input gate, and (U_g, V_g), b_g are input weights, and the bias of intermediate cell state, respectively.

The new cell state C_t may be updated by the output of the forget gate, input gate, and intermediate cell state.

$$C_t = C_{t-1} \cdot f_t + i_t \cdot g_t \quad (14)$$

where "." represents dot product.

Finally, the output of the LSTM network (h_t) can be computed from the output gate O_t and *tanh* function.

$$o_t = \sigma[U_o h_{t-1} + V_o x_t + b_o] \quad (15)$$

$$h_t = o_t \cdot tanh(c_t) \quad (16)$$

where (U_o, V_o) and b_o are the input weights and bias of the output gate.

5. Experimental setup

Two types of bearing vibration data (severity defect data originated from seeded defects and bearing accelerated life data) are used for validating the proposed methodology. The seeded defect data are used for verifying fault classification accuracy, and bearing accelerated life data are used for health indicator construction for bearing RUL prediction. The details of the bearing experimental rigs and datasets are given below:

5.1. Case 1: Bearing seeded defect data

An experimental test rig is designed and developed for bearing vibration data generation, as shown in Fig. 4. The seeded defect data acquired from the bearing test rig have been used to validate the proposed fault classification methodology. The experimental rig consists of a shaft, two test bearings at each end of the shaft, and two load bearings at the center. The load of 1470 N is applied on the load bearings, and the shaft connected to the 3-phase induction motor runs at a speed of 1200 rpm. A single-point circular defect of size 0.6 mm is introduced on the inner race, outer race, and ball (shown in

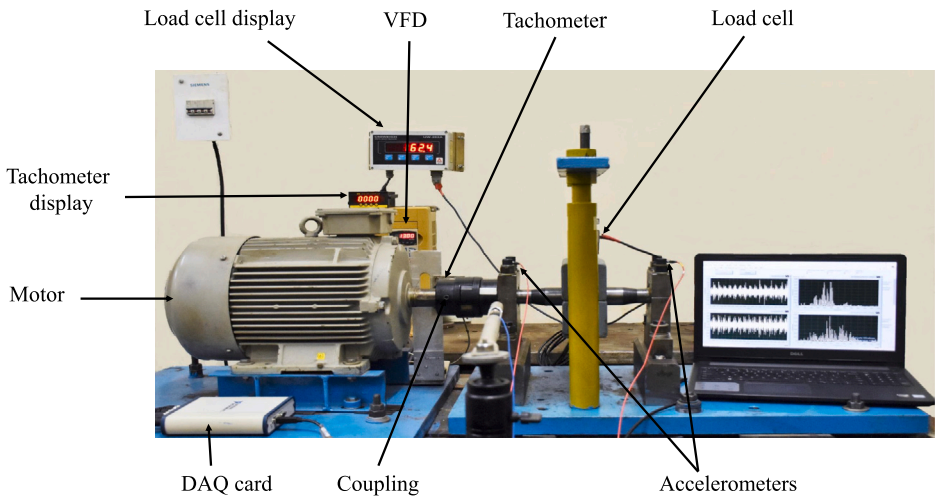


Fig. 4. Bearing experimental test rig.

Fig. 5). These defects are created using electric discharge machining. Two Dytran accelerometers (3055D3 type) are attached to the two test bearings, and NI-USB 4432 national instruments data acquisition card (the resolution of A/D converter is 24 bit) is used to extract vibration data. The vibration data samples of 10-second duration are acquired at every one minute with a sampling frequency of 51.2 kHz (one-second sample file contains 51200 data points). Eight bearing vibration data sets are acquired: one healthy bearing, three single defects (inner race, outer race, and ball defect), and four combined defects (inner race and ball defect, ball and outer race defect, inner race and outer race defect, and combined three defects). The time-domain bearing vibration signal corresponding to healthy, ball defect, inner race defect, and outer race defect are represented in Fig. 6. It may be noted that the amplitude of the healthy bearing vibration signal is minimum, and it has the minimum number of impulses. In defective bearings, a high number of impulses may be observed. For the same defective size, the vibration magnitude of the inner race defect is highest, which is on the expected lines.

5.2. Case 2: Bearing accelerated life test data

The vibration data available at the Center for Intelligent Maintenance Systems, University of Cincinnati [45] are used as validation datasets to validate the algorithm of finding the best health indicator for the bearing prognosis. The vibration datasets were extracted from a run-to-failure bearing test conducted on a specially designed experimental setup. The vibration data files are collected every 10 min with a sampling rate of 20 kHz (one data file consists of 20480 data points).

6. Results and discussion

6.1. Fault diagnosis

The vibration data acquired from the bearing experimental setup (case 1) are used for fault diagnosis. Fifty sample points from each of eight bearing conditions are utilized for validation. First, the vibration data are filtered with an EMD-based filter [35], and random noise present in the acquired data is removed. Multiple signal processing techniques are then applied to the filtered vibration signal, and seventy-two original features are extracted, as mentioned in Section 2. Some of these features contain redundant information, which may reduce the classification performance, cause overfitting, increase complexity and computational time. A Random Forest-based feature selection method is used to eliminate redundant features. The Random Forest algorithm is trained with 60% of the data, and 40% of the data are used for testing. The number of decision trees used is 1000 to obtain the best results, and the Gini index is used to rank the seventy-two original features. A subset of 16 features (STD, RMS, Kurtosis, Peak Factor, Clearance Factor, Shape Factor, RMS*Kurtosis, FFT(BPFI), DWT(Quantile (0.25)), DWT(Quantile (0.75)), EMD(SV1), EMD(SV2), EMD(SV3), EMD(SE1), EMD(SE2), and EMD(SE3)) is selected from the original feature set, providing the maximum accuracy. First, the Random Forest algorithm is trained and tested with seventy-two features. The accuracy of the model with all features obtained is 99.1%. Next, the model is trained and tested with selected features. The selected subset of features increases accuracy from 99.1% to 100%, and time consumption is reduced by 10%. From these results, it is noted that the time-domain features and EMD-based features carry higher weightage than others. The frequency-domain features and wavelet transform-based features showed poor performance. The selected subset of features is used for further bearing fault classification study.

6.1.1. Fault diagnosis of single defects

In this case, a healthy bearing dataset and three defective bearing datasets (ball, inner race, and outer race defects) are used for validation. The fourteen dimensionality reduction techniques are used to extract 2D fault features. First, the selected 16 features are normalized by the min–max normalization method using Eq. (17). Then, the normalized features are fused by proposed techniques, and the first two dimension features that exhibited the maximum variance are chosen for the fault classification. The input parameters are not required for linear techniques except the LPP. In the LPP technique, the nearest neighbors (k) are 20 given as the input parameter. The input parameters given in the nonlinear dimensional techniques are: kernel width $\sigma = 100$ in the KPCA method, $k = 60$ in the ISOMAP, the LLE, and the LTSA methods, $k = 10$ and $\sigma = 200$ in the LE method, number of steps $t = 60$ and $\sigma = 10$ in the DM, and perplexity is 100 in the t-SNE method. In the PCA, the first two eigenvalues exhibited the maximum cumulative proportion variation equal to 0.882 (88.2%). Therefore, the first two dimensional features are chosen for fault classification. The reconstructed ICA algorithm is used in the ICA technique. The feature normalization is done in the following manner:

$$X_{ni} = \frac{X_i - \min(X_i)}{\max(X_i) - \min(X_i)} \tag{17}$$

where X_i is ith feature, and X_{ni} normalized ith feature.

The two-dimensional features are given as input to calculate the BD between fault classes. The BD is calculated between each of the two fault clusters. Six BD values are obtained from six combinations of four bearing fault classes. The separation capability index is calculated by taking the mean of all obtained BD values. The higher value of BD

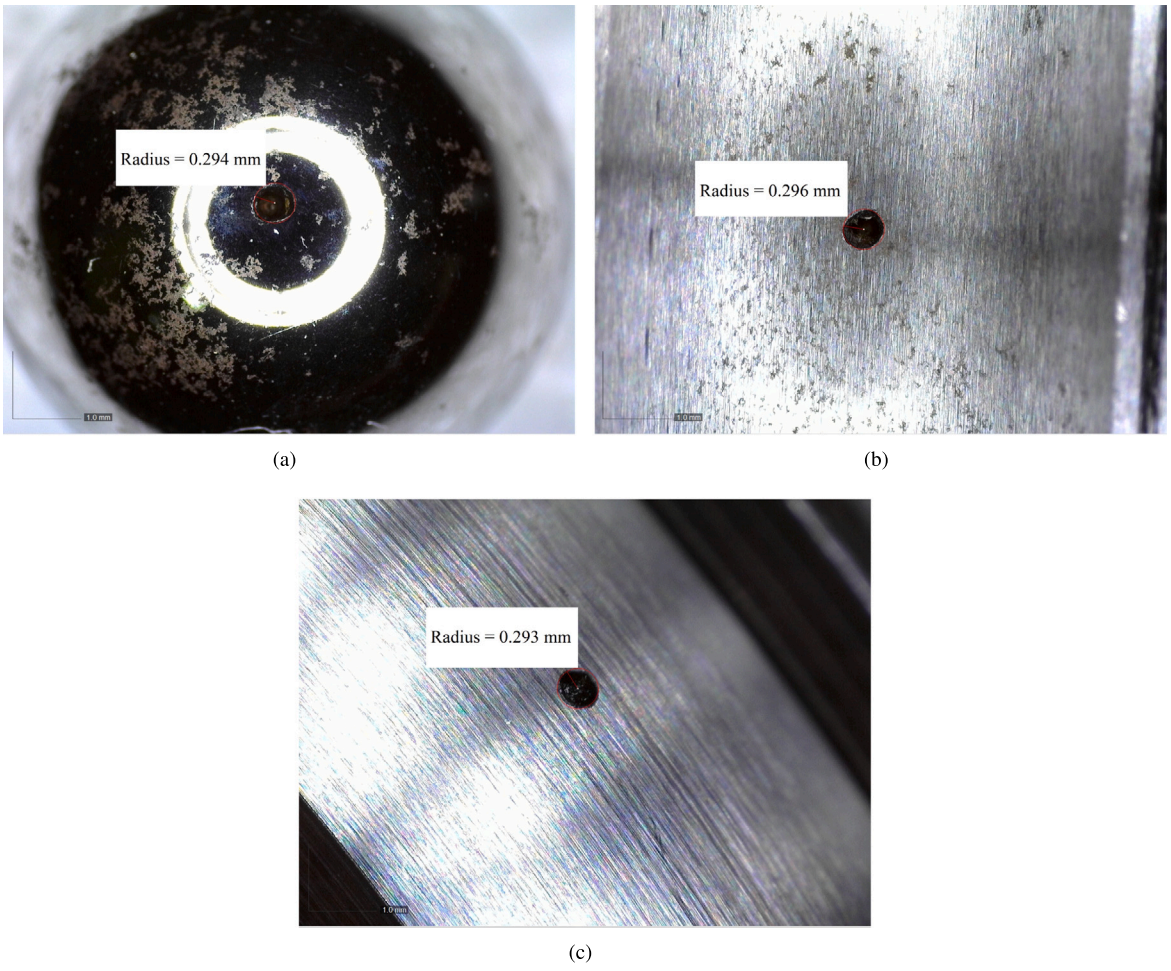


Fig. 5. Defective bearings: (a) Ball defect, (b) Inner race defect, (c) Outer race defect.

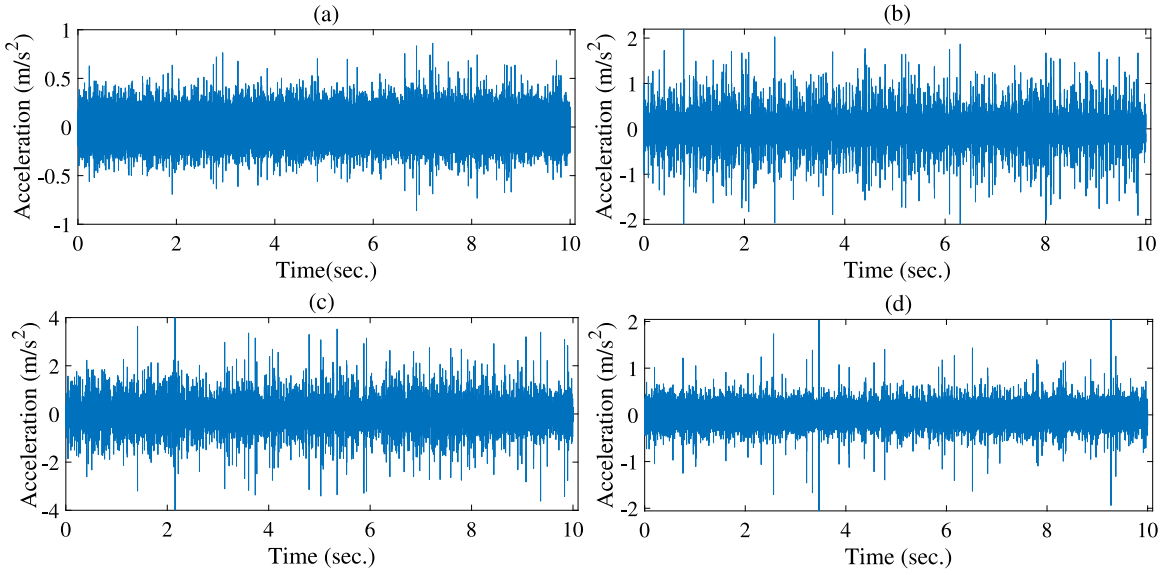


Fig. 6. Measured vibration signals: (a) Healthy, (b) Ball defect, (c) Inner race defect, (d) Outer race defect.

represents the excellent separation of the bearing faults. The BD-based index is calculated for fourteen techniques, and the results are listed in Table 3. The LDA technique yielded the highest BD value (52.2), followed by the t-SNE method (45.9). The two-dimensional features extracted from the dimensionality reduction techniques are inputted to multi-class SVM to compute classification accuracy. Three types of kernels (linear, polynomial (poly), and Radial Basis Function (RBF) kernel) are used in the SVM model. The multi-class SVM is trained and tested for finding fault classification accuracy (in percentage) of the dimensionality reduction techniques, and the results are given in

Table 3
Separation capability and classification accuracy of proposed and other methods for single defects.

Method	PCA	ICA	MD	FA	LDA	CCA	LPP	KPCA	ISOMAP	LLE	LE	DM	LTSA	t-SNE
BD	19.3	45.6	12.5	16.0	**52.2**	19.7	40.7	27.8	41.1	16.5	17.2	16.9	22.8	45.9
SVM (linear)	100	97.5	97.5	91.5	**100**	100	98	95	100	99	100	100	100	100
SVM (poly)	99.5	98	97	96.5	**100**	100	98.5	93.5	100	99.5	99.5	99.5	99.5	99.5
SVM (RBF)	100	97.5	97.5	97	**100**	99.5	99	93.5	99.5	99.5	100	100	100	100
Naive Bayes	100	97	97	91	99.8	99.5	98	92	99	99	99.5	99	99.3	99.5
KNN	99.5	97	97	93.5	99.8	99.5	98	93.5	99.5	99.2	99.5	99.5	99.5	95.5

Table 4
Separation capability and classification accuracy of proposed and other methods for combined defects.

Method	PCA	ICA	MD	FA	LDA	CCA	LPP	KPCA	ISOMAP	LLE	LE	DM	LTSA	t-SNE
BD	26.8	26.7	22.7	36.5	**94.9**	25.4	38.3	27.6	46.9	26.5	32.8	26.3	51.3	92.2
SVM (linear)	99	97.5	88.5	98.8	**100**	99	91	99.1	96	98.3	96	98.3	97.5	99.3
SVM (poly)	99.3	97.5	88.3	98.5	99.8	99.3	90.5	99.3	96.3	97.9	97	98.8	98.8	99.3
SVM (RBF)	99.3	97.5	89	98.5	**100**	99.1	90.8	99	95.9	97.8	96.5	99	97.5	99.3
Naive Bayes	99	97.1	84.5	85.5	99.5	99	90.3	99	94.3	97.8	92.8	98.8	96	99
KNN	99.1	97.2	86.5	98.5	99.5	99	89.8	99.1	95.8	97.7	96	98.3	97.3	99.3

Table 3. The table shows that the LDA method performs best for fault classification, with the SVM-based classification accuracy for all three kernels being 100% and with the highest BD value (52.2). The SVM classification accuracy with two kernels resulted in 100% for seven methods (PCA, CCA, ISOMAP, LE, DM, LTSA, and t-SNE). So, the BD value is considered for selecting the second best method. The t-SNE method performs second-best results with the BD value of 45.9. In order to compare the effectiveness of the proposed method, the KNN, and Naive Bayes classifiers are performed for bearing faults classification. Two-dimensional fault features of fourteen data fusion techniques are given input to KNN, and Naive Bayes classifiers, and results are represented in Table 3. The LDA method shows the highest classification accuracy (99.8%) in both the classifiers. In comparison with KNN and Naive Bayes methods, the SVM offers better classification accuracy.

6.1.2. Fault diagnosis with combined faults

Eight bearing condition datasets: one healthy bearing, three single defects (inner race, outer race, and ball defect), and four combined defects (inner race and ball defect, ball and outer race defect, inner race and outer race defect, and combined three defects) are used for demonstration. The selected subset of features is normalized using Eq. (17). Then, these feature are fused by proposed data fusion techniques, and 2D fault features are extracted. The hyper-parameters in the fourteen data fusion techniques are the same as Section 6.1.1. The 2D fault representations (for all fourteen techniques) by plotting dimension one versus dimension two are shown in Figs. 7 and 8.

The 2D plots show the comparative discrimination capabilities of the eight types of bearing conditions. Twenty-eight BD values are calculated from twenty-eight combinations of eight bearing fault classes from 2D features. The separation capability index is computed by taking the mean of all obtained BD values, and the results are listed in Table 4. In this case also, the LDA techniques resulted in the highest BD value (94.9), and the t-SNE method provides second best results, 92.2. The two-dimensional features are given as input to multi-class SVM to compute classification accuracy. The multi-class SVM is trained and tested for finding fault classification accuracy (in percentage) of the dimensionality reduction techniques, and the results are given in Table 4. From the table, it may be noted that the LDA method performs best for fault classification with the SVM-based classification accuracy with two (linear and RBF) kernels being 100% and with the highest BD value (94.9). In comparison with KNN and Naive Bayes methods, the SVM offers better classification accuracy.

6.2. Prognosis

Three accelerated life test bearing data sets are utilized for the validation, and details of these data sets are given in Table 5. The

Table 5
Bearing accelerated life test datasets details.

Data set no.	No. of data files	Type of defect
1	984	Outer race
2	2156	Inner race
3	2156	Roller element

vibration data is filtered with an EMD-based filtering approach [35]. Then, seventy-two features (as mentioned in Section 2) are extracted from each of three vibration data sets by applying multiple signal processing techniques. The Random Forest algorithm is used to reduce the dimensionality of the feature set. The Random Forest algorithm is trained with 60% of the data, and 40% of the data are used for testing, and the number of decision trees used is 1000. The best subset of features selected from three data sets improves the accuracy in comparison to all features. In data set no. 1, the Random Forest algorithm selects fifteen features that increase accuracy from 99.2% to 99.7%, and time consumption is reduced by 10.2%. In the data set no. 2, thirteen features are selected with an accuracy increase from 96.5% to 98.4%, and the processing time reduced by 21.3%. Fifteen features are chosen in the data set no. 3, with an increase of accuracy from 98.0% to 98.7%, and 10% of time consumption is reduced. The details of the selected features, increase in accuracy (in percentage), reduction in computational time (in percentage) from each data are given in Table 6. Fig. 9(a) shows the health indicator obtained by fusing seventy-two original features, and Fig. 9(b) shows the health indicator obtained by fusing selected fifteen features (for dataset no. 1). Heavy fluctuation due to overfitting in the healthy state and failure stages are observed in Fig. 9(a). The health indicator in Fig. 9(b) is smooth in the healthy state, and a clear trend may be observed in the later failure stages. Therefore, overfitting is reduced by using selected features for data fusion.

The selected features are normalized using Eq. (17) and then fused by the data fusion techniques. The highest variance dimension feature is chosen as a health indicator. In the PCA, the maximum proportion of variance is explained by the first eigenvalue, 0.981 (98.1%). Therefore, the first principal component is chosen as the health indicator. For linear and non-linear techniques, three health indicators are computed from three data sets. The resulting plots (health indicator value versus time in multiple of 10 min) are shown in Figs. 10 and 11. The health indicators represent the degradation trend of bearing life. A good health indicator should be selected for prognosis model building to improve the bearing RUL prediction accuracy. The population of three health indicators of each technique is used to compute four prognosis health indicator selection attributes mentioned in Section 4.3. A fitness function is formulated by calculating the weighted sum of four attributes

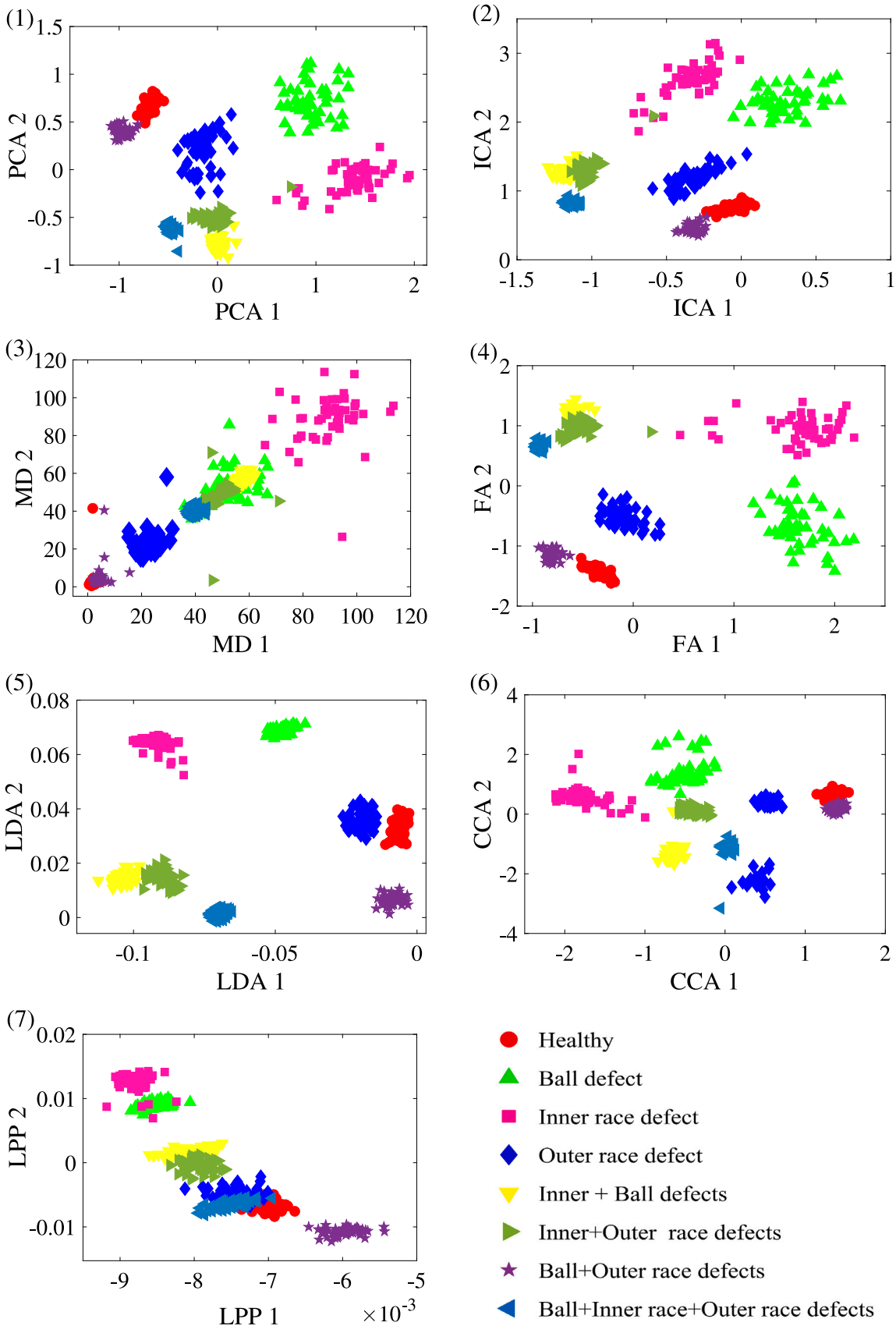


Fig. 7. 2D fault features of linear dimensionality reduction techniques: (1) PCA, (2) ICA, (3) MD, (4) FA, (5) LDA, (6) CCA, (7) LPP.

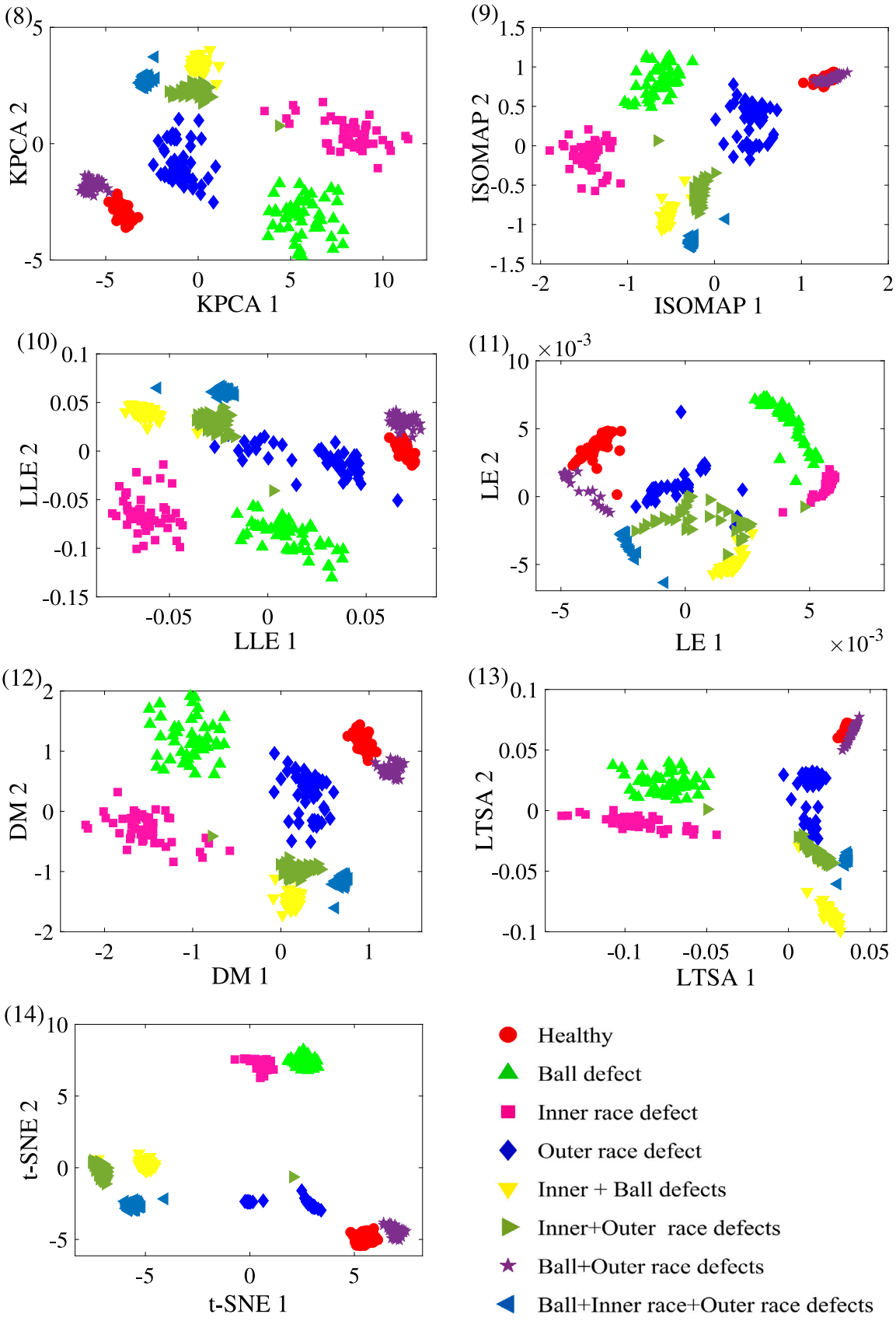


Fig. 8. 2D fault features of non-linear dimensionality reduction techniques: (8) KPCA, (9) ISOMAP, (10) LLE, (11) LE, (12) DM, (13) LTSA, (14) t-SNE.

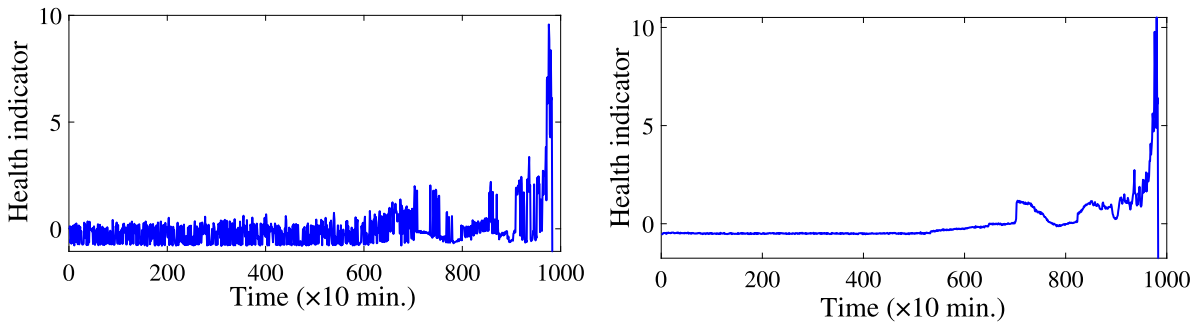


Fig. 9. Fusion-based health indicators: (a) using seventy-two features, (b) using selected 15 features.

Table 6
Selected subset of features.

Observation	Data set no.1		Data set no.2		Data set no.3	
Selected features	RMS*Kurtosis	RMS	Shape factor	RMS	Shape factor	RMS
	Shape factor	STD	Impulse factor	Peak factor	Clearance factor	Peak factor
	$DWT_{RMS*Kurtosis}$	DWT_{RMS}	RMS*Kurtosis	STD	RMS*Kurtosis	STD
	$DWT_{HFRT(BPFO)}$	DWT_{Asinh}	$DWT_{Quantile\ (0.25)}$	DWT_{RMS}	$CWT_{Kurtosis}$	$CWT_{Peak\ factor}$
	$DWT_{Quantile\ (0.25)}$	EMD_{SV1}	$CWT_{Quantile\ (0.75)}$	EMD_{SV1}	$CWT_{RMS*Kurtosis}$	EMD_{SV1}
	$DWT_{Quantile\ (0.75)}$	EMD_{SV2}	$DWT_{Quantile\ (0.5)}$	EMD_{SV2}	$CWT_{Quantile\ (0.75)}$	EMD_{SV2}
	EMD_{SE1}	EMD_{SE2}	EMD_{SV3}		$CWT_{Quantile\ (0.25)}$	EMD_{SE2}
	EMD_{SE3}				EMD_{SE3}	
No. of features selected	15		13		15	
Accuracy in % (with selected features)	99.7		98.4		98.7	
Accuracy in % (with total features)	99.2		96.5		98.0	
Reduction in computational time in % (with selected features)	10.2		21.3		10.0	

to construct a suitability metric used to compare health indicators and select the best one. Initially, weights of all attributes were calculated by an optimization method, the Genetic algorithm (computed using MATLAB®). This algorithm allocates the weights such that maximum value to the attribute, which is the highest value, and zero value to minimum value attribute. In this case, weights obtained by Genetic algorithm are, for monotonicity (w_M) = 0.000, prognosability (w_P) = 0.033, trendability (w_T) = 0.800, and SNR (w_S) = 0.160. The weight of the monotonicity is zero, which means this attribute is neglected from the fitness function due to the objective being purely mathematical. However, monotonicity is important in measuring the suitability of prognosis health indicators. Therefore, to solve this problem, a simple mathematical equation (Eq. (18)) is formulated for calculating the weights of four attributes. The weights of the attributes are calculated by dividing the average of fourteen values of each attribute by the sum of the averages of four attributes. The weights of the attributes are computed by the following equation:

$$w_j = \frac{A_j}{U} \tag{18}$$

where w_j is the weight of the attribute, $A_j[j = M, P, T, S]$ is the average of fourteen values (respective to fourteen techniques) of an attribute, and U is the sum of the averages of four attributes ($U = A_M + A_P + A_T + A_S$).

The weights of attributes obtained by above equation are $w_M = 0.096, w_P = 0.238, w_T = 0.352$, and $w_S = 0.311$. The values of four attributes of fourteen techniques and their fitness values (H) are shown in Table 7. From the table, it is noticed that the PCA, the ICA, and the LTSA have the same fitness value (around 0.55), and the LLE has a slightly higher value (0.563). Therefore, these four health indicators are best suited for RUL prediction.

The bearing RUL has been predicted using all health indicators of the data set number 1. The LSTM technique is used to build the prognosis model for predicting the RUL of the bearing. The data fusion-based health indicators are given input to the LSTM network. The network trained with 70% of the data, and 30% of the data is used for testing. The LSTM model is trained with 200 hidden layers with a learning rate of 0.005. The LSTM method predicts the following one step and then updates the predicted values at each time step. The model predicts the remaining life of the bearing at each step in the training data. The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are computed to evaluate the RUL performance of the data fusion-based LSTM method, and the results are listed in Table 8. The performance of the LSTM prognosis model largely depends on the input health indicator. The health indicator with a high fitness value may provide the best RUL results. From Table 8, it is concluded that the LTSA-based RUL prediction provides the best results with the minimum RMSE and MAE values, 4.05 and 3.49, respectively. The LLE-based RUL prediction shows the second-best performance. The PCA and the ICA-based prediction also show promising results. To show the effectiveness of the proposed LSTM-based prognosis method, the results are compared with the 1d-CNN-based method. The hyperparameters set in the 1D-CNN algorithm are the number of hidden units 100, initial learning rate 0.01, dropout rate 0.5, and the maximum number of epochs is given as 100. The algorithm is trained with 70% of the data, and 30% is used for testing. The RMSE and MAE values are used for comparing the two prognosis methods. The LSTM-based model shows better results compared with the CNN. The RUL (multiple of 10 min) results of bearing using the LTSA and the LLE-based methods are demonstrated in Fig. 12.

7. Summary and conclusions

A holistic framework is proposed for data fusion based bearing fault diagnosis and prognosis. The efficient feature extraction based on fourteen data fusion techniques has been carried out for bearing faults classification and RUL prediction. The best health indicator and 2D fault features are identified by comparing fourteen dimensionality reduction techniques.

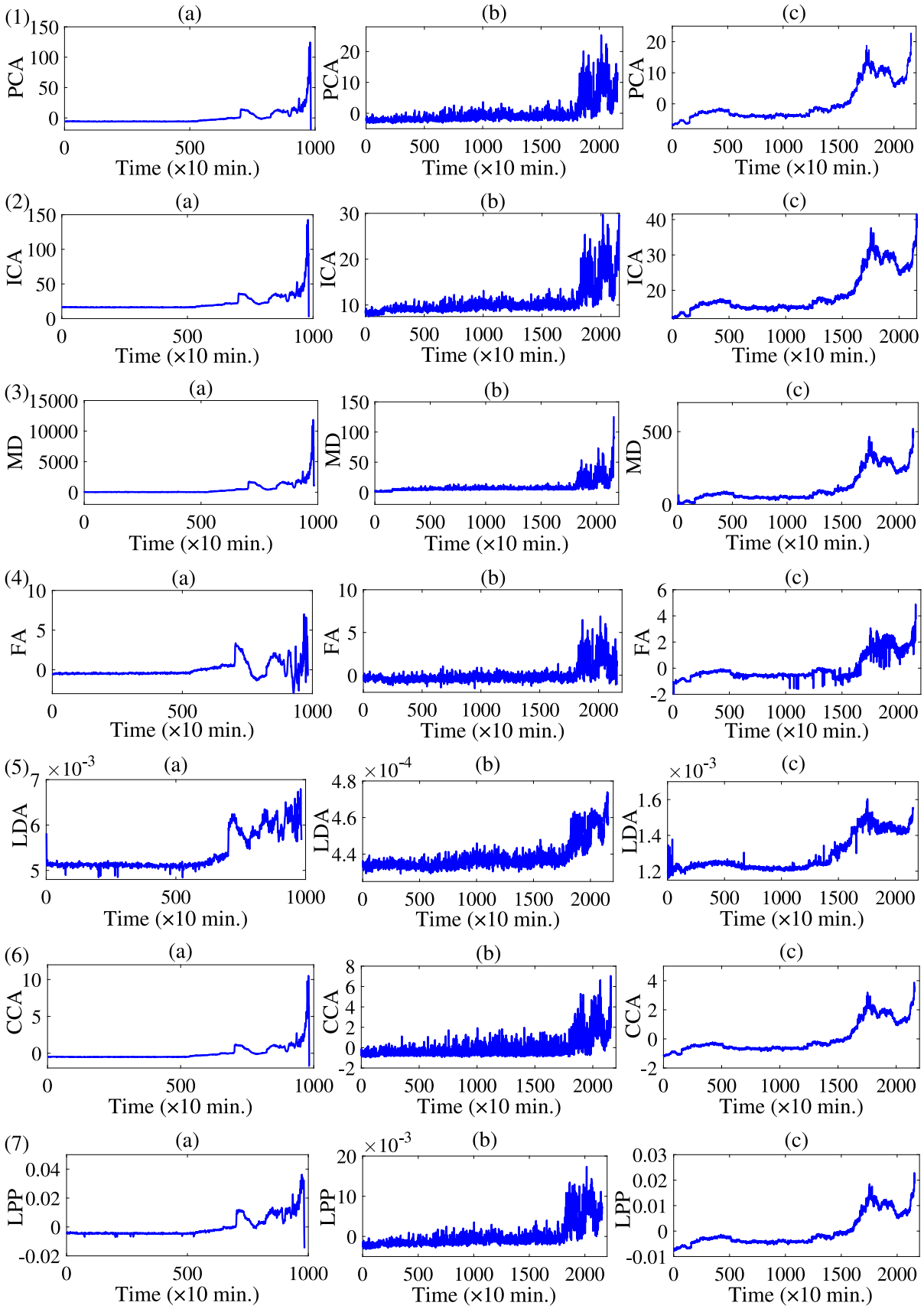


Fig. 10. Health indicators of linear dimensionality reduction techniques (x-axis is time in multiple of 10 min) [(a) Data set no. 1, (b) Data set no. 2, (c) Data set no. 3]: (1) PCA, (2) ICA, (3) MD, (4) FA, (5) LDA, (6) CCA, (7) LPP.

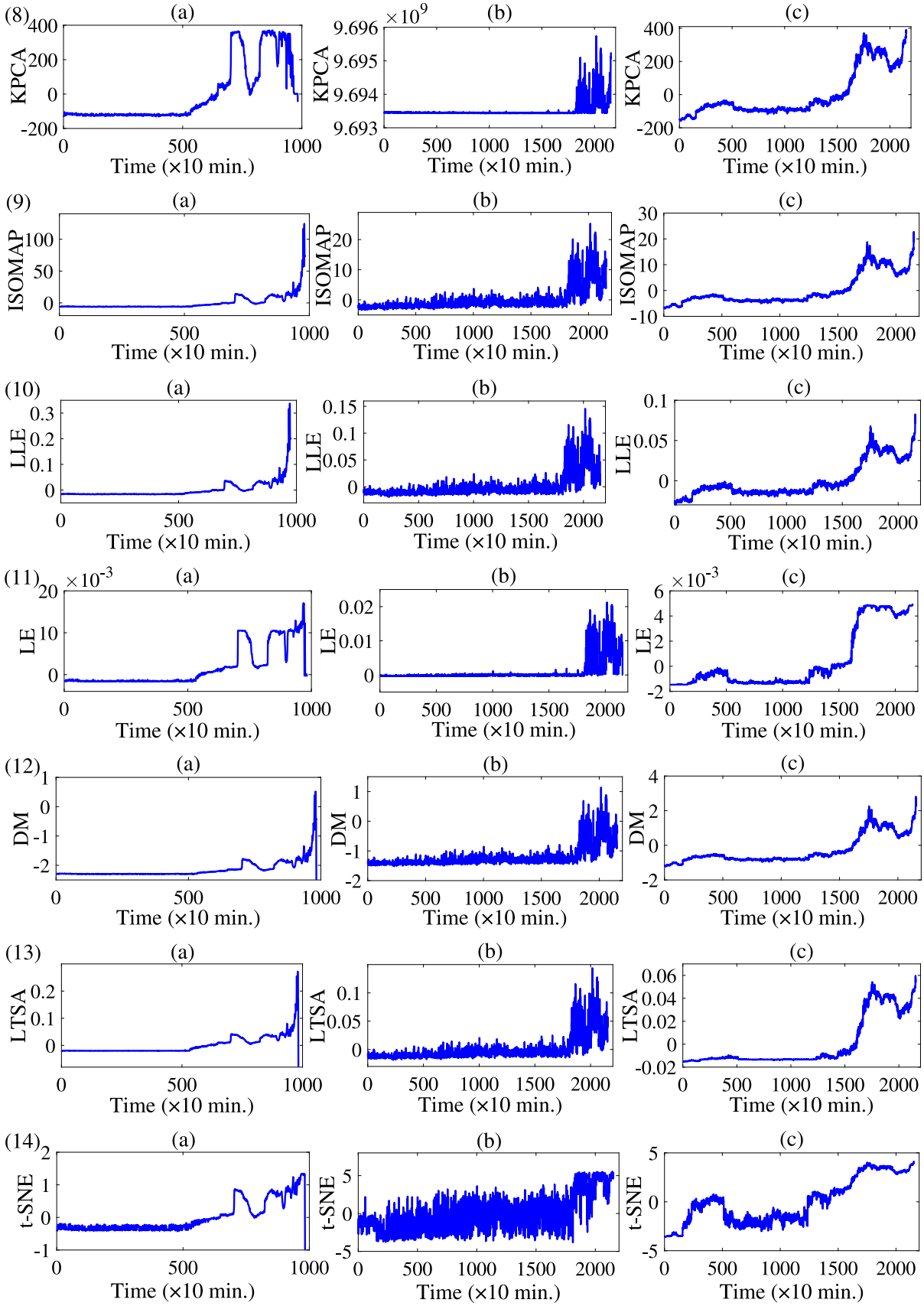


Fig. 11. Health indicators of non-linear dimensionality reduction techniques (x-axis is time in multiple of 10 min) [(a) Data set no. 1, (b) Data set no. 2, (c) Data set no. 3]: (8) KPCA, (9) ISOMAP, (10) LLE, (11) LE, (12) DM, (13) LTSA, (14) t-SNE.

Table 7
Attributes and fitness values of data fusion techniques.

S. no.	Technique	Monotonicity	Prognosability	Trendability	SNR	Fitness value
1	PCA	0.168	0.383	0.703	0.629	0.551
2	ICA	0.176	0.429	0.614	0.714	0.557
3	MD	0.182	0.474	0.617	0.579	0.529
4	FA	0.106	0.202	0.315	0.261	0.251
5	LDA	0.163	0.603	0.747	0.394	0.546
6	CCA	0.189	0.352	0.639	0.539	0.496
7	LPP	0.21	0.41	0.67	0.573	0.533
8	KPCA	0.155	0.781	0.471	0.463	0.512
9	ISOMAP	0.168	0.407	0.606	0.62	0.52
10	LLE	0.173	0.424	0.736	0.593	**0.563**
11	LE	0.171	0.222	0.661	0.695	0.519
12	DM	0.168	0.371	0.628	0.617	0.519
13	LTSA	0.151	0.434	0.71	0.608	**0.558**
14	t-SNE	0.217	0.415	0.619	0.44	0.475

Table 8
The RUL results of proposed method and other method.

Observation		PCA	ICA	MD	FA	LDA	CCA	LPP	KPCA	ISOMAP	LLE	LE	DM	LTSA	t-SNE
LSTM	RMSE	4.65	4.8	7.45	9.05	60.63	6.35	6.37	13.51	5.55	**4.17**	26.4	6.15	**4.05**	24.84
	MAE	4.02	4.24	7.08	8.93	43.62	5.90	5.81	8.06	5.12	**3.75**	23.2	5.72	**3.49**	18.04
CNN	RMSE	5.45	5.64	7.82	4.79	61.93	6.73	7.65	13.52	6.48	5.67	23.98	6.87	5.36	25.84
	MAE	5.38	5.57	7.72	4.7	45.79	5.65	5.58	8.11	5.4	5.6	23.97	5.87	5.27	18.1

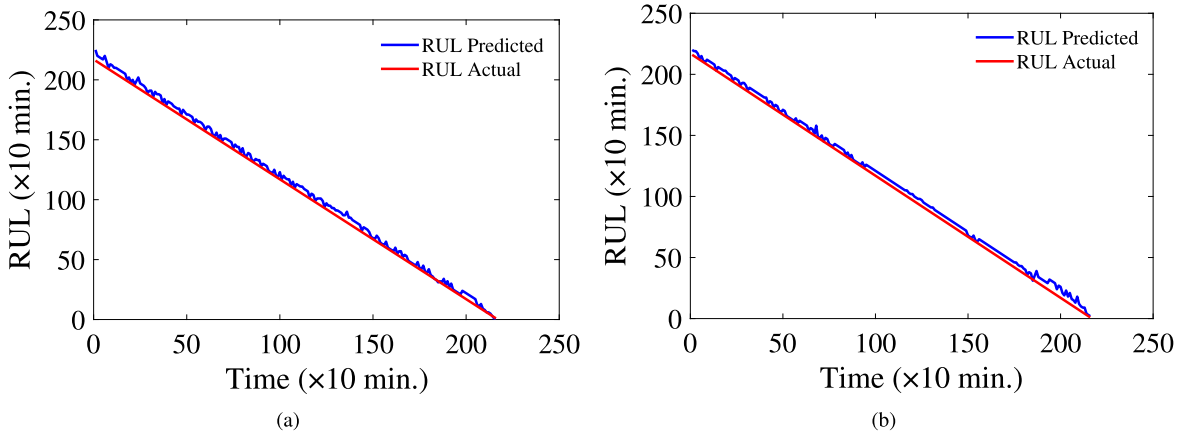


Fig. 12. The LSTM-based RUL results of bearing (x-axis is time in multiple of 10 min): (a) LTSA-based method. (b) LLE-based method.

1. Seventy-two original features are extracted from bearing vibration signals using various signal processing techniques. The relevant features are selected from the extracted features by the Random Forest method. The accuracy is increased by feature selection from 99.1% to 100% in seeded defect data (case 1), and in case 2, the accuracy increased from 99.2% to 99.7% for data set no. 1, 96.5% to 98.4% for data set no. 2, and 98.0% to 98.7% for data set no. 3. The computational time reduction in all datasets in between 10% to 21%.
2. The selected features are fused by fourteen dimensionality reduction techniques, and significant lower-dimensional features are extracted. The high variance first dimension feature of each method is used as the prognosis health indicator, and the first two dimensional features are used for fault classification.
3. The BD value is calculated between the fault clusters, and classification of different fault types is carried out by the SVM method. Bearing single defects and combined defects are classified successfully. The BD value and classification accuracy based comparison is made between the techniques. The LDA shows its superiority with the highest BD value (52.2) and 100% classification accuracy for single fault classification. For combined fault classification also, the LDA provides the best results with a BD value of 94.9 and 100% classification accuracy. The superiority of SVM is shown by comparing it with the KNN and the Naive Bayes methods.
4. The fitness value is computed for fourteen dimensionality reduction techniques based health indicators to select the best one. From the fitness value results, it may be noted that the LTSA and the LLE are the best techniques for constructing a significant health indicator for building a prognosis model of rolling element bearing. The bearing RUL predicted by the LSTM method using the LTSA and the LLE-based health indicators provides accurate RUL results. The LSTM offers precise results compared with the CNN method.

The proposed methodology provides data fusion-based feature extraction to improve the bearing fault diagnosis and prognosis accuracy. It provides a framework for selecting the best data fusion technique for extracting 2D fault features and the health indicator. In the real-world industry application, data fusion-based feature extraction may improve fault diagnosis and prognosis accuracy of machine elements (e.g., gears, motors, shafts, turbine blades, etc.). Further research may include automatic input parameters selection in the dimensionality reduction techniques.

CRediT authorship contribution statement

Sandaram Buchaiah: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Writing – original draft. **Piyush Shakya:** Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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